Deepit Bhatia

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Research interests

Focused on bridging the gaps between theoretical and applied infectious disease epidemiology to study how disease elimination can be achieved through vaccination in human and livestock diseases. Experienced in data analysis and stakeholder engagement in public health.

Education

2021 – 2026 **The Pennsylvania State University** – University Park, PA, USA
PhD in Biology
Supervisor: Dr. Matthew Ferrari

2013 – 2015 **University of Toronto** – Toronto, ON, Canada
MPH in Epidemiology
Practicum Supervisors: Dr. Jeff Kwong, Dr. Sarah Richmond, Dr. Andrew Howard

2009 – 2013 **McGill University** – Montréal, QC, Canada
BSc (Hons.) in Environment, Ecological Determinants of Health

Research experience

Jan 2021 – Jul 2026 **Penn State University**
Supervisor: Dr. Matthew Ferrari
I studied how we can understand disease dynamics, and consequently, plan for elimination of vaccine-preventable diseases such as measles and foot-and-mouth disease.

Jan 2015 – Apr 2015 **ICES**
Supervisor: Dr. Jeff Kwong
I studied the effect of changes in routine Pap tests on chlamydia and gonorrhea incidence in Ontario, Canada.

May 2014 – Dec 2014 **The Hospital for Sick Children**
Supervisors: Dr. Andrew Howard, Dr. Sarah Richmond
I investigated the impact of bike lanes on reducing injury to cyclists in Toronto, Ontario, Canada.

May 2012 – Aug 2012 **McGill University Health Centre**
Supervisor: Dr. Madhukar Pai
I consulted with stakeholders in tuberculosis testing in India to discuss the feasibility of leveraging existing logistics systems to conduct centralized sputum sample collection and diagnosis.

Relevant work experience

Apr 2020 – Oct 2020
York Region Public Health (Epidemiologist) – Vaughan, ON, Canada
- Managed outbreaks in 30+ long-term care homes by assessing resident and staff status and recommending courses of action to prevent spread of COVID-19
- Collaborated with IT staff to securely share health data between schools/long-term care facilities and the public health unit

Jan 2020 – Apr 2020
Public Health Agency of Canada (Contract Researcher) – Toronto, ON, Canada
- Created a systematic framework to identify future research priorities within the Agency related to the impact of climate change on infectious disease risk in Canada through interviews with Agency staff and a literature review

Jan 2017 – Sept 2019
BlueDot Inc. (Epidemiologist/Project Coordinator) – Toronto, ON, Canada
- Developed spatial methodology to assess the spread of vector-borne disease in Florida at a census tract level using local mobility data
- Produced daily reports for the Public Health Agency of Canada assessing the risk of importation of ongoing infectious disease events to Canada
- Trained senior and technical officials in the Association of Southeast Asian Nations on tools to increase regional biosecurity capacity in infectious disease outbreak preparedness and response

Apr 2015 – Dec 2016
ICES (Analyst) – Toronto, ON, Canada
- Linked medication, billing, demographic, and social determinant data from administrative databases to create project-specific analysis datasets
- Conducted statistical analyses using individual-level health administrative data for cancer and infectious diseases studies

Honours and scholarships

2025 Peter J. Hudson Best Student Paper Award, $100 (Center for Infectious Disease Dynamics, Penn State University)

2025 CIDD Travel Award, $500 (Center for Infectious Disease Dynamics, Penn State University)

2024 Finalist (1 of 10), Penn State University 3 Minute Thesis Competition

2023 CIDD Travel Award, $500 (Center for Infectious Disease Dynamics, Penn State University)

2021 University Graduate Fellowship, $30,000 USD/year, 2 years (Penn State University)

2021 Homer F. Braddock Scholarship in Biology, Chemistry, and Physics, $5,000 USD (Penn State University)

2020 Ontario Veterinary College Scholarship–Declined, $25,000 CAD/year, 3 years (University of Guelph)

2017 TalentEdge Internship Award, $35,000 CAD (Ontario Centres for Excellence)

2016 Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) Travel Award, $1,000 USD (University of Washington)

Publications and Manuscripts

2026 *Preprint:* **Bhatia, D.**, McKee, A., & Ferrari, M. J. (2026). Characterizing the relative strength of metapopulation dynamics in measles persistence. *medRxiv*, 2026.10.24.26366357. https://doi.org/10.64898/2026.10.24.26366357.

2025 **Bhatia, D.**, Crowcroft, N., Antoni, S., Danovaro-Holliday, M. C., Bose, A. S., Minta, A., Masresha, B., & Ferrari, M. J. (2025). Prediction of subnational-level vaccination coverage estimates using routine surveillance data and survey data. *Vaccine*, 60, 127277. https://doi.org/10.1016/j.vaccine.2025.127277.

2020 Lapointe-Shaw, L., Rader, B., Astley, C. M., Hawkins, J. B., **Bhatia, D.**, Schatten, W. J., Lee, T. C., Liu, J. J., Ivers, N. M., & Stall, N. M. (2020). Web and phone-based COVID-19 syndromic surveillance in Canada: A cross-sectional study. *PloS One*, 15(10), e0239886.

2020 Giannakeas, V., **Bhatia, D.**, Warkentin, M. T., Bogoch, I. I., & Stall, N. M. (2020). Estimating the Maximum Capacity of COVID-19 Cases Manageable per Day Given a Health Care System's Constrained Resources. *Annals of Internal Medicine*, 173(5), 407–410. https://doi.org/10.7326/M20-1169

2020 Tuite, A. R., **Bhatia, D.**, Moineddin, R., Bogoch, I. I., Watts, A. G., & Khan, K. (2020). Global trends in air travel: Implications for connectivity and resilience to infectious disease threats. *Journal of Travel Medicine*, 27(4), taaa070.

2019 Grubaugh, N. D., Saraf, S., Gangavarapu, K., Watts, A., Tan, A. L., Oidtman, R. J., Ladner, J. T., Oliveira, G., Matteson, N. L., Kraemer, M. U. G., Vogels, C. B. F., Hentoff, A., **Bhatia, D.**, Stanek, D., Scott, B., Landis, V., Stryker, I., Cone, M. R., Kopp, E. W., … Andersen, K. G. (2019). Travel Surveillance and Genomics Uncover a Hidden Zika Outbreak during the Waning Epidemic. *Cell*, 178(5), 1057-1071.e11. https://doi.org/10.1016/j.cell.2019.07.018

2019 O'Reilly, K. M., Lowe, R., Edmunds, W. J., Mayaud, P., Kucharski, A., Eggo, R. M., Funk, S., **Bhatia, D.**, Khan, K., Kraemer, M. U. G., Wilder-Smith, A., Rodrigues, L. C., Brasil, P., Massad, E., Jaenisch, T., Cauchemez, S., Brady, O. J., & Yakob, L. (2018). Projecting the end of the Zika virus epidemic in Latin America: A modelling analysis. *BMC Medicine*, 16(1), 180. https://doi.org/10.1186/s12916-018-1158-8

2018 Bogoch, I. I., Maxim, T., Acosta, H., **Bhatia, D.**, Chen, S., Huber, C., Janes, A., Yong, J. H., Thomas, A., & Kraemer, M. U. (2018). Potential plague exportation from Madagascar via international air travel.*The Lancet Infectious Diseases*, 18(3), 247–248.

2018 Tuite, A. R., Thomas-Bachli, A., Acosta, H., **Bhatia, D.**, Huber, C., Petrasek, K., Watts, A., Yong, J. H., Bogoch, I. I., & Khan, K. (2018). Infectious disease implications of large-scale migration of Venezuelan nationals. textitJournal of Travel Medicine, 25(1), tay077.

2017 Naimer, M. S., Kwong, J. C., **Bhatia, D.**, Moineddin, R., Whelan, M., Campitelli, M. A., Macdonald, L., Lofters, A., Tuite, A., & Bogler, T. (2017). The effect of changes in cervical cancer screening guidelines on chlamydia testing. The Annals of Family Medicine, 15(4), 329–334.

2017 **Bhatia, D.**, Richmond, S. A., Loo, C. J., Rothman, L., Macarthur, C., & Howard, A. (2016). Examining the impact of cycle lanes on cyclist-motor vehicle collisions in the city of Toronto. *Journal of Transport & Health*, 3(4), 523–528.

Manuscripts in Preparation

Studying the decline in seroprevalence after the introduction of Foot-and-mouth disease vaccination in India. In collaboration with Dr. Glen Guyver-Fletcher, Dr. Matthew Ferrari, and Dr. Mike Tildesley.

Tracking progress to measles elimination using routinely collected subnational surveillance data: a case study from Nigeria. Led by Dr. Indrajit Ghosh, in collaboration with Dr. Matt Ferrari and Dr. Amy Winter.

Modelling Foot-and-Mouth Disease in India. Led by Dr. Guyver-Fletcher, in collaboration with Dr. Matt Ferrari and Dr. Mike Tildesley.

Predicting the performance of supplementary immunization activities. Manuscript in preparation.

Invited Talks

Jun 2025 **Invited talk**: Prediction of subnational-level vaccination coverage estimates using routine surveillance data and survey data
World Health Organization, with representatives from measles-endemic countries

Nov 2018 **Invited talk, with Dr. Ashleigh Tuite**: Measuring infectious diseases globally
Cancer Care Ontario Discussion Club

Mar 2018 **Invited talk**: John Snow and Cholera
Pivotal Inc

Peer-reviewed Talks and Posters

Nov 2025 **Talk**: Higher vaccination coverage can lead to lower seroprevalence through differential waning and pulsed vaccination
Northeast Ecology and Evolution of Infectious Diseases 2025, Penn State

Jun 2025 **Poster**: Forecasting the impact of immunization campaigns
Ecology and Evolution of Infectious Diseases 2025, Notre Dame

May 2025 **Poster**: Forecasting the impact of immunization campaigns
Ecological Forecasting Initiative, Virginia Tech

Jun 2024 **Poster**: The impact of stray cattle on foot-and- mouth disease control in India: a modeling study
Ecology and Evolution of Infectious Diseases 2024, Stanford University

Jan 2024 **Talk**: Understanding the role of metapopulation persistence in measles dynamics
Binghamton University Biology Graduate Student Symposium

Nov 2023 **Poster**: Identifying local and regional persistence in measles dynamics in connected populations
EPIDEMICS9, Bologna, Italy

May 2023 **Poster**: Identifying local and regional persistence in measles dynamics in connected populations
Ecology and Evolution of Infectious Diseases 2023, Penn State University

June 2022 **Poster**: Measles surveillance data are a better predictor of vaccination coverage than administrative coverage data
Ecology and Evolution of Infectious Diseases 2022, Emory University

Teaching experience

Summer 2024 **Teaching assistant, Mathematical Models of Infectious Disease**: SISMID, Emory University

Spring 2024 **Teaching assistant, BIOL 230**: Cellular and Molecular Biology, Penn State University

Spring 2023 **Teaching assistant, BIOL 419/519**: Ecological and Environmental Problem Solving, Penn State University

Summer 2022 **Teaching assistant, Mathematical Models of Infectious Disease**: SISMID, University of Washington

Guest Lectures

Sep 2024 **Lecture**: Predator-prey dynamics
BIOL220M, Penn State University

Sep 2023 **Lecture** (with Dr. Ariel Greiner): Logistic Growth
BIOL 220, Penn State University

Mar 2023 **Lecture**: Compartmental Models
BIOL 419/519, Penn State University

Service and Leadership

2025 – Present Steering Committee Member, Ecological Forecasting Initiative (Canadian chapter)

2025 – Present Peer-reviewer, Vaccine

2025 – 2026 Organizing Committee, 2026 Ecological Forecasting Initiative Annual Conference

2021 – 2025 Assisted with student recruitment for prospective CIDD graduate students at PSU

2024 Spoke on a graduate student and faculty panel on mental health in academia (Invited Panelist)

2024 Organized a 3 Minute Thesis showcase for students at the Center for Infectious Disease Dynamics

2023 – 2024 Center for Infectious Disease Dynamics Graduate Student Organization (President)

2023 oSTEM at Penn State (Invited Panelist)

2023 Undergraduate Science Exhibit (Judge)

2023 Ecology and Evolution of Infectious Disease (EEID) Conference (Volunteer)

2022 – 2023 Center for Infectious Disease Dynamics Graduate Student Organization (Social Chair)

Mentorship

2024 Career and academic mentor to a first-year MPH student at the Dalla Lana School of Public Health, University of Toronto

Technical skills

Programming languages
Proficient in: R (*tidyverse*, *ggplot2*, *leaflet*, *Shiny*)
Familiar with: Python, SAS

Software
RStudio, GitHub, Office, QGIS

Languages
English (fluent), Hindi (native language), French (B)

Other interests

Bicycling, hiking, cooking, foraging, plant and fungus identification, fibre arts.

References

References available upon request.